

Rushikesh Pawar

pawar.rush@northeastern.edu | +1(617)-505-8797 | [LinkedIn](#) | [GitHub](#)

EDUCATION

Northeastern University | Boston, MA
Master of Science in Computer Science
CDAC IACSD | Pune, India
PG Diploma in Big Data Analytics

Jan 2025 – May 2027

Aug 2021 – May 2022

SKILLS

Programming languages: Python, Java, C, C++, Embedded C, MATLAB.

ML/DS libraries/skills: Pandas, Numpy, OpenCV, Scikit-learn, matplotlib, TensorFlow, Keras, PyTorch, Cuda, NLTK.

Tools & Technologies: Docker, AWS, Git, GitHub, Power BI, CARLA Simulator, MATLAB, Linux/Unix, SQL, MongoDB.

WORK EXPERIENCE

Data Scientist – FEV | India

May 2022 – Oct 2024

- Led a cross-functional team of 5 to analyze large-scale EV battery testing datasets, creating predictive models for RUL, SOH, and SOC and performing hyper-parameter optimization, **improving estimation accuracy by 18%**.
- Created a digital twin of EV batteries for Tata Motors using deep learning models, building a real-time monitoring dashboard that **reduced fault detection time by 40%** and improved testing efficiency.
- Designed and prototyped custom computer vision systems for traffic sign classification, traffic-light recognition, and lane departure warning in Indian road scenarios, **reducing false negatives by over 50%** in controlled tests.
- Implemented ALiVA, an automated annotation pipeline for images, video, and LiDAR using multi modal approach to increase the accuracy, **cutting manual labeling time by 40%**.
- Generated diverse road scenarios using CARLA simulation to support perception model testing, reducing real-world data generation costs by 60% and **saving approximately 2 months** of testing. Enabled rapid preliminary algorithm validation.

Data Analyst – Comperv Consultancy | India

Apr 2020 – May 2021

- Analyzed customer reviews and user preference datasets to derive actionable insights that informed targeted marketing strategies, directly contributing to a **20% revenue increase** year-over-year.
- Managed and cleaned terabytes of enterprise data using SQL and Python, ensuring **99%+ data integrity** by identifying and resolving inconsistencies, missing values, and errors across multiple data sources.
- Conducted comprehensive statistical analyses on customer behavior patterns, uncovering key trends and correlations that optimized product positioning and marketing campaigns.
- Built unified analytics dashboard using Power BI for 4 different projects with real-time KPI tracking and integrated community hub, enabling cross-functional collaboration and **reducing status meetings by 40%**.
- Executed end-to-end data analytics workflows including data collection from databases and surveys, data wrangling, visualization with Matplotlib, and presentation of actionable recommendations to stakeholders.

RESEARCH PAPERS

- Towards Automation of Reference Data Generation for ADAS/AD Function Development – AliVA Framework
- Comparative Analysis of GenAI Models for Extension and Validation of EV Battery Characterization Data

PROJECTS

Smart Attendance System

Apr 2022

- Developed an automated classroom attendance system leveraging real-time computer vision to eliminate manual roll-call processes, reducing attendance tracking time by 90% and minimizing human error.
- Implemented YOLOv5/CNN-based object detection and facial recognition algorithms to accurately identify and verify students, achieving 95%+ recognition accuracy in varied lighting conditions.

Smart Prescription Management System

Jun 2021

- Developed full-stack web application enabling digital prescription creation with automated PDF generation and WhatsApp API integration, reducing prescription processing time by 60% and eliminating paper-based errors.
- Built analytics dashboard visualizing prescription trends, patient visit patterns, and medication analytics, providing data-driven insights that improved clinic operational efficiency by 40%.
- Designed secure database architecture for HIPAA-compliant storage of patient records and prescription history, enabling instant retrieval of complete medical histories and streamlining clinic operations.